Priyansha

Mechanical Engineer — Robotics Engineer
priyansha04@gmail.com — +91 9998624881
LinkedIn — Portfolio

Work Experience

Robotics Intern June 2024 - August 2024
TIH IHub Drishti - Jodhpur

- Implemented visual SLAM-based mapping and compared Monocular, Monocular-inertial, Stereo, and Stereo-inertial vSLAM
- Robotics Operating System, Gazebo Simulation, Embodied AI
- Navigation and path planning of Clearpath Husky Robot UGV

Intern May 2023 - September 2023
Insights Autotech Consultants - Remote

- Market Research, Mechanical Design, and relevant research
- Sales strategy, Production Planning, Layout and Process Planning

Education

Malaviya National Institute of Technology Jaipur *2022-2026*
Bachelor of Technology in Mechanical Engineering, Honours in Robotics and Automation CGPA: 7.98/10
Strong skills in operations research, control systems, robot vision, and design of mechanical systems
Proficient in ML, Python, MATLAB, ABAQUS, Witness Simulation, LabView, AutoCAD, Fusion360

Navrachana Higher Secondary School (CBSE) Vadodara *2014-2021*
High School Diploma
12th Board: 92.8% — 10th Board: 95.5%
TCS IT Wiz Ahmedabad Regional Runner Up, 2018
IIT Bombay Techfest Earth Matters finalist

Skills

- Mechanical Design
- Machine Learning
- Finite Element Analysis (ABAQUS, Ansys)
- CAD Software (SolidWorks, AutoCAD, Fusion360)
- Product Design
- Robot Operating System (ROS2)
- Python, Computer Vision
- Simulation (Gazebo, CoppeliaSim, LabView)
- GNU Octave, MATLAB
- Arduino, Control Systems

Projects

Design and Control of Robot Arm *01/2025 – Current*

- Tools Used: ROS2, Computer Vision, Inverse Kinematics
- Designed and controlled a 3D-printed robot arm with servo motors and a camera
- Implemented ROS2 for communication, computer vision for navigation, and inverse kinematics for precise movement control

Design and Analysis of Automobile Transmission System *2024*

- Tools Used: Fusion360 and ABAQUS
- Conducted design and finite element analysis of an automobile transmission system

E-Yantra Robotics Competition *09/2023 – 01/2024*

- Simulated a self-balancing bike robot using a rotary inverted pendulum for balancing
- Implemented PID control for stabilization
- Software Used: CoppeliaSim, GNU Octave, Arduino, Python

Data-Driven Study of Mental Health of University Students in India *09/2023 – Current*

- Used Machine Learning to assess the mental health of university students in MNIT campus
- Based on the PGWBI scale, working under guidance of Dr. Gunjan Soni with psychologist Dr. Tvishi Sharma

Food Waste Management System *12/2023*

- Developed a hostel food wastage management system using weight sensors and Arduino
- Integrated data acquisition system for tracking food wastage in messes
- Sensors Used: Fingerprint Sensor, Load Cell, HX711 weight sensor module
- Software Used: Arduino, MS Excel, PLX-DAQ